

Iteration in Programming

while loops

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Topics list

1. Repetition in Programming – Intro to looping
2. Use of loops (while loops).

Recap: Boolean conditions

- A boolean condition is an expression that evaluates to either true or false e.g.

`mouseX < 50`

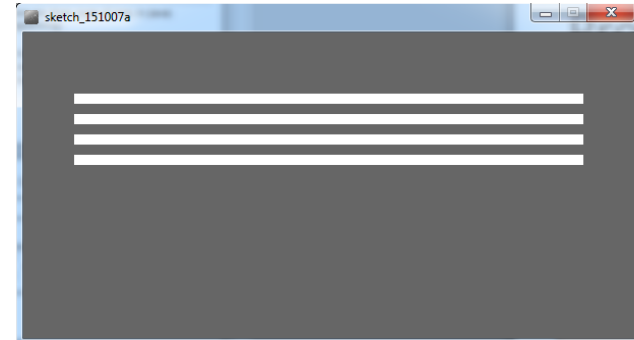
- Boolean conditions can be used to control:
 - Selection i.e. if statements and
 - Iteration i.e. loops (we will look at these now).

Repetition in Programming

- Computers are very good at repetition.
- Example:
 - **calculate pay** for 1000 employees.
 - You should use the same **calculate pay** algorithm 1000 times.
 - You don't write the **calculate pay** algorithm 1000 times; instead you include it in a loop.

Form of loop

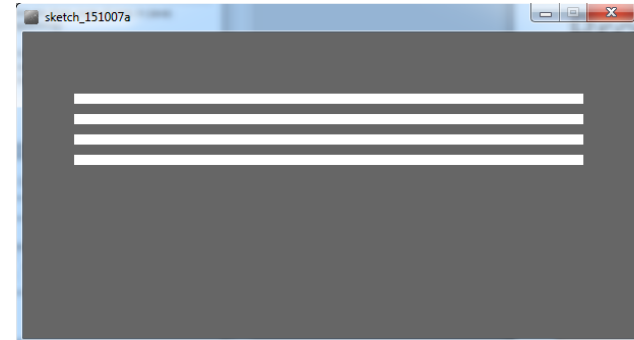
- Draw a rectangle 4 times that has a gap of 10 pixels between each one.



Form of loop

- Draw a rectangle 4 times that has a gap of 10 pixels between each one.
 - Without loop:

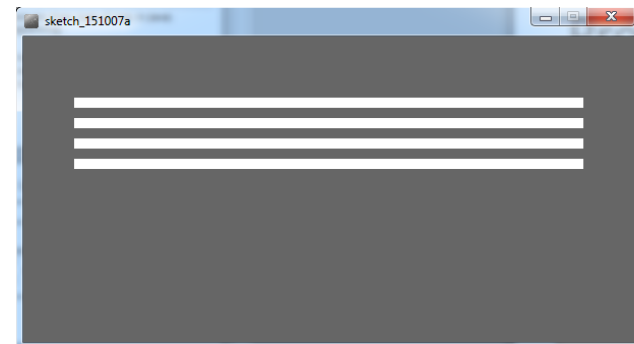
```
rect(50, 60, 500, 10);  
rect(50, 80, 500, 10);  
rect(50, 100, 500, 10);  
rect(50, 120, 500, 10);
```



Form of loop

- Draw a rectangle 4 times that has a gap of 10 pixels between each one.
 - With a loop:
 - do this 4 times
(adding 20 onto the yCoordinate variable each time).

```
rect(50, yCoordinate, 500, 10);
```

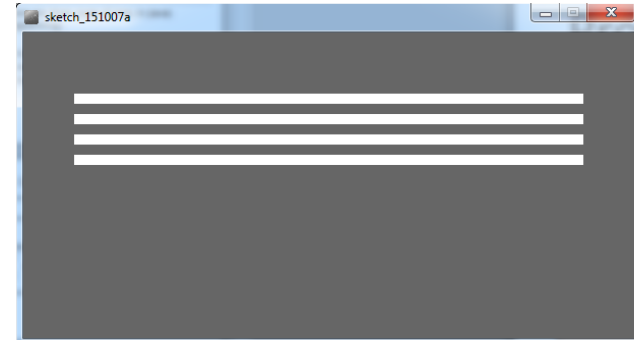


Form of loop

- Draw a rectangle 4 times that has a gap of 10 pixels between each one.
 - With a loop:
 - do this 4 times
(adding 20 onto the yCoordinate variable each time).

```
rect(50, yCoordinate, 500, 10);
```

- *We will learn a little more about loops and then we will write the code to solve this problem.*



Topics list

1. Repetition in Programming – Intro to looping
2. Use of loops (while loops).

Loops in Programming

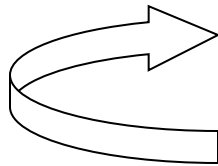
- There are three types of loop in (Java) programming:
 - **while** loops
 - **for** loops
 - **do while** loops

while loop pseudo code

General form of a while loop

while keyword

boolean condition



```
while(loop condition) {  
    loop body  
}
```

Statements to be repeated

Pseudo-code expression of the actions of
a while loop

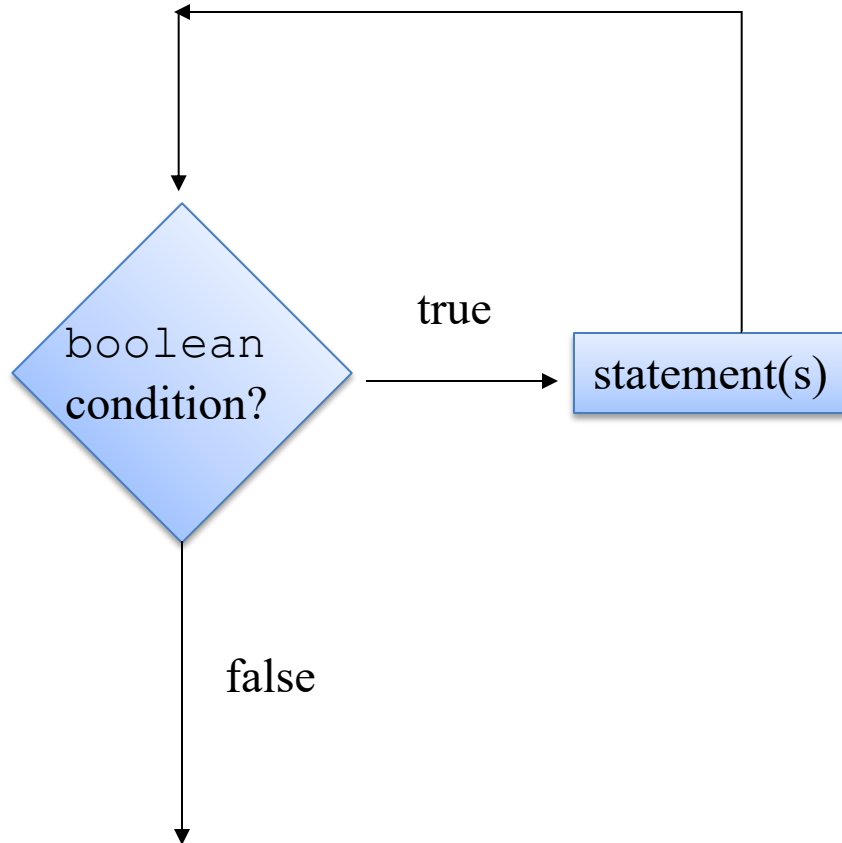
while we wish to continue, do the things in the loop body

Construction of while loop

```
Declare and initialise loop control variable (LCV)
while(boolean condition based on LCV is true)
{
    "do the job to be repeated"
    "update the LCV"
}
```

This structure should always be used

while loop Flowchart



```
int yCoordinate = 60;
```

```
int i = 0;           //i is the LCV
```

```
while(i < 4)
```

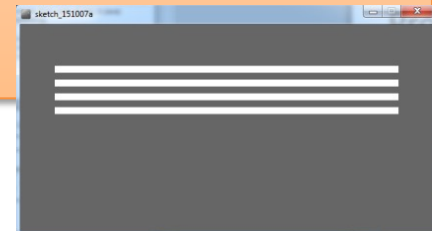
```
{
```

```
    rect(50, yCoordinate, 500, 10);
```

```
    yCoordinate += 20;
```

```
    i++;
```

```
}
```

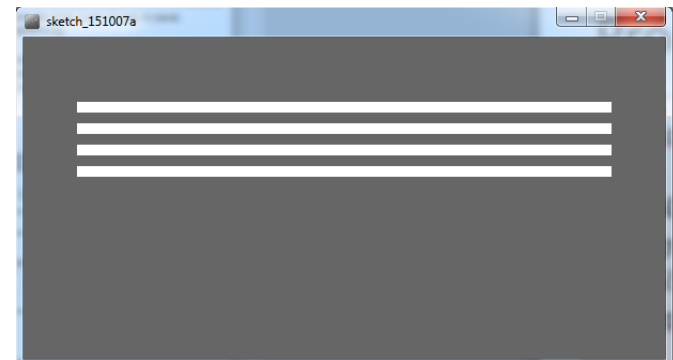


Processing Example 4.5

```
int yCoordinate = 60;

size(600, 300);
background(102);
fill(255);
noStroke();

int i = 0;
while(i < 4)
{
    rect(50, yCoordinate, 500, 10);
    yCoordinate += 20;
    i++;
}
```



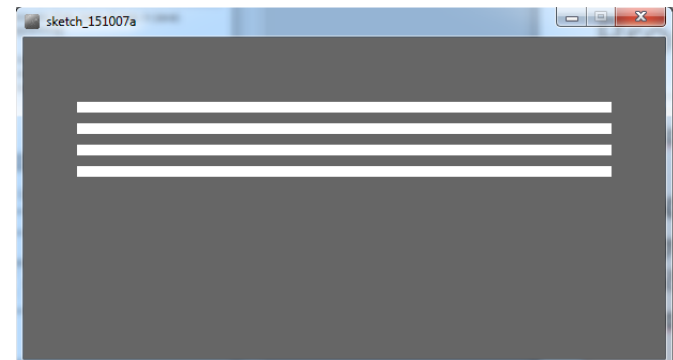
Processing Example 4.5

```
int yCoordinate = 60;

size(600, 300);
background(102);
fill(255);
noStroke();

int i = 0;
while(i < 4)
{
    rect(50, yCoordinate, 500, 10);
    yCoordinate += 20;
    i++;
}
```

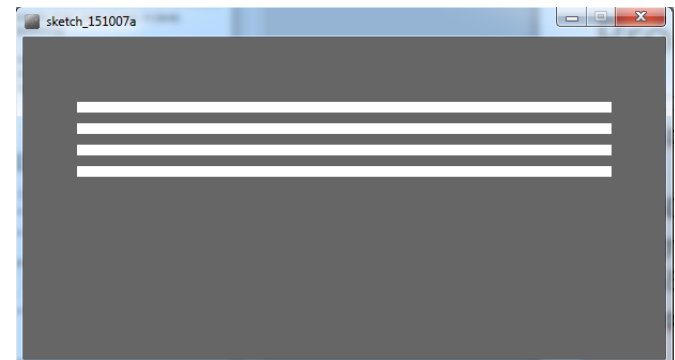
Q: Could we remove the **yCoordinate** variable and rework the code to still produce the four lines using the while loop?



Processing Example 4.6

```
size(600, 300);  
background(102);  
fill(255);  
noStroke();  
  
int i = 60;  
while(i <= 120)  
{  
    rect(50, i, 500, 10);  
    i += 20;  
}
```

A: Yes. Here is the solution with *no yCoordinate* variable.



Some Study Exercises

This basic while loop,
produces this output.

```
int i = 1;

while (i <=5)
{
    println("Hello World");
    i++;
}
```



```
Hello World
Hello World
Hello World
Hello World
Hello World
```

Some Study Exercises

1. Change the code so that “Hello World” is printed out 10 times.
2. Change the code so that the numbers from 1 to 10 (inclusive) are printed out, one line at a time.
3. Change the code so that the numbers from 10 to 1 are printed out.

Questions?

